CONCEPTUAL PROPOSAL · NOT AN IMPLEMENTATION REPORT

Project World: A Research Framework for a Voluntary, Resource-Aware Credit-Clearing Network

Conceptual Architecture, Governance Risks, and an Empirical Evaluation Agenda

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ABSTRACT

Project World is a proposal for a voluntary credit-clearing network through which participating households, businesses, and cooperatives could exchange goods and services using non-fiat credits while separately recording selected physical resource flows. This paper develops the idea as a research proposal rather than as a validated economic system. It distinguishes three instruments that are often conflated: physical resource accounts, transferable claims on specified deliveries, and mutual-credit settlement balances. This distinction matters because a measured quantity of energy or food is not, by itself, a generally redeemable payment instrument.

The paper presents a conceptual architecture, accounting model, governance requirements, risk analysis, and staged evaluation plan. It does not assume that blockchain or artificial intelligence is necessary; those technologies would need to demonstrate measurable benefits over simpler systems. It also does not claim that resource-linked credits guarantee price stability, ecological sustainability, equality, peace, or independence from existing institutions. The central research question is whether a voluntary, resource-aware credit network can improve exchange and resource visibility for a defined community without imposing excessive transaction costs, financial risk, exclusion, or ecological harm.

Keywords: complementary currency; mutual credit; resource accounting; community exchange; ecological economics; distributed ledger; governance

1 Introduction

Project World asks whether a community or network of organizations could use its own credit arrangements to settle some exchanges while keeping clearer records of the resources involved. The idea is motivated by concern about resource use, waste, and environmental impacts. The United Nations Environment Programme's Global Resources Outlook 2024, for example, documents rising material extraction and associated environmental pressures; it does not, however, establish that changing the form of money would solve those problems.

The proposal therefore makes a narrower, testable claim:

> A voluntary credit network may help participants coordinate certain local exchanges and make selected resource flows more visible, but its economic and environmental effects must be measured rather than assumed.

The terms credit system and moneyless economy should not be treated as equivalent. If a credit is used to quote prices, pay wages, and buy a range of goods, it is money-like in function, even if it is not issued by a central bank or pegged to a national currency. Monetary functions commonly include serving as a unit of account, a medium of exchange, and a store of value.

Project World is therefore framed here as a non-fiat credit system, not as an economy without money in the functional sense.

2 Scope and Contributions

This paper offers four contributions:

1. Conceptual clarification: It separates physical resource measurement from payment and credit issuance.
2. System design: It outlines a voluntary network using delivery claims, mutual-credit balances, or both.

3. Risk analysis: It identifies financial, technical, ecological, and governance risks that would need to be addressed.
4. Evaluation agenda: It proposes research questions and measures for a simulation or small-scale pilot.

This paper does NOT claim that Project World:

- can replace national or global monetary systems;
- guarantees zero inflation or removes debt;
- automatically reduces resource use or environmental damage;
- makes war impossible;
- allows an AI to allocate resources fairly or accurately;
- has been implemented, tested, or shown to outperform existing arrangements.

These are questions for research, not conclusions.

3 Background and Related Work

3.1 Resource use and environmental accounting

Resource use has environmental consequences that are not captured by a single measure such as energy consumption. UNEP's global assessment discusses material extraction, emissions, waste, and related impacts across economic systems. This supports the case for tracking relevant physical indicators, but not for treating any single resource measure as a complete account of economic or ecological value.

Project World should therefore distinguish economic settlement from environmental measurement. A credit ledger can record who paid whom; it cannot, on its own, establish that a transaction is environmentally sustainable.

3.2 Complementary currencies and credit networks

Complementary currency research already examines accounting systems created by non-bank actors to support exchange within defined trading networks. Related work discusses how special-purpose currencies interact with wider monetary systems, including through their liquidity, convertibility, and acceptance.

Adoption is also a research question, not a given. A case study of eight local currency initiatives in Switzerland reported difficulty reaching a critical mass of regular users and businesses. That finding is specific to the studied schemes, but it highlights a practical issue for Project World: a credit is useful only where participants can spend or redeem it.

3.3 Money, liquidity, and price discovery

A credit network must specify how prices are expressed, how payment becomes final, how participants trust each other's balances, and what happens when members cannot meet obligations. The BIS identifies a common unit of account, trust in settlement, liquidity, and interoperability as important properties of effective monetary arrangements.

A centralized AI allocator should not be treated as if it can know every participant's needs, constraints, and local circumstances. The problem of using dispersed and incomplete knowledge in economic coordination has a long history in economic thought. Project World should preserve participants' ability to set offers and make choices, rather than assuming an algorithm can determine the right value for all goods.

3.4 Distributed ledgers and physical-world evidence

A blockchain or other shared ledger can help preserve a record after information has been entered. It cannot by itself prove that a physical event occurred as reported. Smart contracts that rely on external information face an “oracle” problem: they need a process for obtaining and validating real-world data.

For Project World, this matters whenever a credit is said to represent a physical output — for example, energy produced, food delivered, or ecological restoration completed.

4 Definitions

Term	Definition in this paper
Resource account	A record of a measured physical quantity or service, such as kilowatt-hours delivered or kilograms of a specified food. It is not automatically a payment instrument.

Delivery credit	A claim by a holder to receive a specified good or service from a named issuer under defined conditions.
Mutual-credit balance	A participant's positive or negative account position within a network. A transaction increases one participant's balance and decreases another's, subject to agreed limits.
Impact indicator	A separate environmental measure, such as greenhouse-gas emissions, material use, water use, or land impacts.
Credit-clearing network	The rules, accounts, and processes through which participating members record and settle exchanges.
Non-fiat	Not denominated in, or guaranteed to be redeemable for, a national currency. This does not mean that the network has no money-like functions or no external dependencies.

Credits such as Energy Credits, Food Credits, and Transit Credits need precise definitions before they can be used. For example:

- “1 Energy Credit” could mean one kilowatt-hour delivered by a named supplier at a specified location and time — not one kilowatt-hour available anywhere at any time.
- “1 Food Credit” would need to specify the food type, quantity, quality, delivery period, and redemption terms.
- “1 Transit Credit” would need to specify the service, route or zone, and applicable conditions.
- “Leisure Credit” is not a physical measurement. It may be a useful spending category, but it should not be described as objectively backed by a single resource.

A broad label such as “Bio-Credit” is not sufficient to establish what a holder can claim.

5 Proposed System Architecture

Project World should be designed as a network with separate layers rather than as one token that is assumed to represent all value.

5.1 Layers

1. Participants: Households, businesses, cooperatives, suppliers, and service providers.
2. Resource and service records: Offers, inventories, delivery records, and impact data.
3. Verification: Invoices, receipts, meter data, inspections, or other evidence, with a defined process for corrections and disputes.
4. Marketplace: Participants publish offers, accept offers, and set exchange terms.
5. Credit clearing: The system records payments, balances, credit limits, and redemption of claims.
6. Governance and audit: Members approve rules, review disputes, monitor risks, and change the system under published procedures.



Figure 1. Proposed transaction and review cycle. This is a proposed design diagram. It does not represent a deployed system.

5.2 Two forms of credit

Project World could test two different forms of credit:

- Delivery claims: A supplier issues a credit that can be redeemed for a defined good or service. The issuer has an obligation to deliver or otherwise resolve the claim.
- Mutual-credit settlement: Members trade through accounts that can hold positive or negative balances within agreed limits. This can support exchanges between people who do not directly barter with one another, but it creates credit risk and requires rules for defaults and imbalances.

These instruments should not be treated as interchangeable. A delivery claim is a promise by an issuer; a mutual-credit balance is an account position in a network.

5.3 Prices and exchange

If Project World uses separate credit types with no common unit, participants must work with exchange ratios or price bundles. That may preserve the purpose of category-specific credits, but it can make pricing and payment more complex.

A second option is to use one community settlement credit for general exchange while separately recording resource accounts and optional delivery claims. This is simpler to price in, but it is money-like by function.

The project should test these alternatives rather than claim that one is objectively correct. Exchange rates should follow transparent, participant-agreed rules. An AI system could optionally suggest matches or flag unusual records, but it should not silently set prices, decide what people are allowed to buy, or control access to essential services.

5.4 Technology choice

Blockchain should be treated as an implementation option, not a premise. A signed database or federated ledger may be cheaper, easier to correct, and easier to govern for a small pilot. A distributed ledger should be selected only if it provides a demonstrated advantage for the participants' trust and coordination requirements. Current monetary-system analysis also emphasizes the risks of fragmented networks and the importance of interoperability.

6 Accounting Model

6.1 Physical resource accounts

For resource r, at location ℓ, during period t, define:

- I(r, ℓ, t): recorded inventory or deliverable capacity;
- P(r, ℓ, t): verified additions or production;
- D(r, ℓ, t): delivered quantity;
- L(r, ℓ, t): losses, spoilage, or other adjustments.

Then:

$$I(r, \ell, t+1) = I(r, \ell, t) + P(r, \ell, t) - D(r, \ell, t) - L(r, \ell, t)$$

A discrepancy between the ledger and a physical inspection is an audit issue, not something a smart contract can automatically resolve.

6.2 Delivery-claim limits

If a delivery credit represents a claim on a specified resource or service, the network should limit outstanding claims to the quantity that can plausibly be delivered under the defined terms:

$$O(r, \ell, \tau) \le A_deliverable(r, \ell, \tau) - R(r, \ell, \tau)$$

where O is the outstanding claim, A_deliverable is audited inventory or contracted capacity for the relevant location and delivery period, and R is any reserve held against disruption or non-delivery.

This rule reduces the risk of issuing more claims than the issuer can meet. It does not eliminate the possibility of supplier failure, inaccurate measurements, or a physical shortage.

6.3 Mutual-credit balances

For a closed mutual-credit network with participant balances b(i), a transaction of q credits can be recorded as:

$$b'(\text{buyer}) = b(\text{buyer}) - q$$
$$b'(\text{seller}) = b(\text{seller}) + q$$

If the system has no external issuer or reserve account, the double-entry rule is:

$$\Sigma_i\, b(i) = 0$$

Each member's balance should have an explicit limit:

$$-L(i) \le b(i) \le U(i)$$

Here, L(i) is the member's permitted negative balance and U(i) is any positive-balance limit. The network must define who sets these limits, how defaults are handled, and whether members can appeal a decision.

A salary paid in mutual credits must be accounted for as a transfer from an existing balance or as a permitted extension of credit with a corresponding debit. It is not automatically “minted” merely because a person has worked.

6.4 Issuance, transfer, and redemption

Payment and redemption are different events:

- A payment usually transfers a balance from buyer to seller.
- A delivery credit may be retired when the promised good or service is redeemed.
- A mutual-credit balance is not necessarily burned when a purchase occurs; it changes account holders' positions.

Consequently, “mint when produced and burn when consumed” should not be presented as a complete monetary model. It describes only one possible voucher or claim mechanism. It does not by itself define wages, prices, credit limits, defaults, or liquidity.

6.5 No guarantee of price stability

A resource-denominated credit does not guarantee stable purchasing power. A fixed energy quantity may be scarce at a particular place or time; food prices and availability can change; and different goods are not interchangeable just because they can be measured physically. A system with multiple credit types also needs a method for expressing relative prices.

A general price index could be created for research purposes, but its basket and weights would be policy choices. The paper does not assume that resource-linked issuance prevents inflation or price volatility.

7 Environmental Safeguards and Governance

7.1 Measure impacts separately

Project World should report environmental indicators separately rather than combine them into a single "Earth Credit." A candidate impact vector could include:

$$m = (\text{GHG emissions},\ \text{material use},\ \text{water use},\ \text{land impact},\ \text{waste})$$

The selection, measurement boundaries, and weighting of these indicators should be documented. A payment record is not evidence of environmental benefit. The system should measure both absolute impact and impact per delivered service, because a lower impact per item does not necessarily mean lower total impact.

7.2 Restoration claims

If Project World later rewards environmental restoration, credits should be issued for verified outcomes rather than simply for an activity such as planting a tree. For climate-related claims, UNFCCC validation and verification standards include requirements such as defining the project boundary, identifying leakage, demonstrating additionality, assessing non-permanence risk, and monitoring outcomes. These provide relevant design precedents, though they do not by themselves establish a general standard for every type of ecological restoration.

7.3 Rights, inclusion, and oversight

A pilot should include:

- voluntary participation and a clear right to exit;
- privacy rules and limits on collecting personal consumption data;
- a dispute process, appeal route, and correction mechanism;
- accessible ways to participate without a smartphone or continuous internet access;
- transparent rules for credit limits, insolvency, and system changes;
- independent review of resource claims and financial records;
- protection against using credits as the sole gatekeeper for essential food, water, healthcare, or shelter.

These are governance requirements, not features that blockchain or AI can guarantee.

8 Potential Advantages and Risks

Potential advantage	Why it may be useful	Risk or limitation	What to measure
Resource visibility	Separate records can show quantities, delivery timing, and selected impacts.	Data may be incomplete, inaccurate, or costly to verify.	Audit discrepancies; measurement cost; coverage of eligible transactions.
Local exchange	A network may help members trade where conventional payment options are limited or costly.	A credit is useful only if enough suppliers accept it.	Acceptance rate; repeat use; share of eligible trade settled in credits.
Purpose-bound support	Restricted claims may direct support to a specified good or service.	Restrictions can be paternalistic, exclusionary, or technically misapplied.	Redemption failures; appeals; access to essential services.
Flexible credit	Mutual credit may permit exchange without requiring every buyer to hold a positive balance first.	Negative balances create default and loss-allocation questions.	Arrears; defaults; balance concentration; limit changes.

Environmental accounting	Separate impact indicators may make resource consequences more visible.	Tracking alone does not ensure lower total impact or fair distribution.	Absolute and per-service impacts versus a baseline.
Shared digital records	A shared ledger may reduce some reconciliation work among independent participants.	A blockchain cannot prove that off-ledger events occurred as reported.	Reconciliation time; oracle errors; disputes; operating cost.

Complementary currency studies discuss potential social and economic roles, while also identifying challenges in governance, connection to wider economic structures, and implementation. The Project World proposal should therefore be evaluated against those existing systems, not described as a proven replacement for them.

9 Research Questions and Hypotheses

Research questions

- RQ1: Can a resource-aware credit network reduce the administrative cost of exchanging goods and services among participating organizations?
- RQ2: Does recording physical resource accounts change purchasing or production decisions, and if so, in what direction?
- RQ3: How do separate resource credits compare with a common community settlement credit in terms of acceptance, price clarity, and exchange friction?
- RQ4: Which governance and verification arrangements best control fraud, default, exclusion, and privacy risk in a small network?
- RQ5: Does the system reduce absolute resource use, or only change the way existing activity is recorded and paid for?

Testable hypotheses

These are hypotheses to test — not findings:

- H1: Separate resource credits may improve the traceability of specified deliveries but may also increase transaction time and conversion friction.
- H2: A common community settlement unit may make pricing and payment easier than a fully multi-credit system, but may make resource-specific constraints less visible to users.
- H3: Adoption will depend on the number and diversity of participating suppliers, the reliability of redemption, and the ease of using credits for routine purchases.
- H4: Any environmental benefit will depend on measured changes in total resource use and sourcing, not simply on the number of credits issued.

10 Evaluation Plan

Phase 1: Protocol specification

Before building software, publish:

- definitions of each credit and the claim it represents;
- issuance, transfer, redemption, and default rules;
- membership and credit-limit rules;
- governance, audit, privacy, and appeal procedures;
- the data needed to estimate environmental impacts.

Phase 2: Simulation and threat analysis

Compare at least two designs:

1. separate resource-specific claims and accounts;
2. a mutual-credit network with resource indicators recorded separately.

The simulation should test different network sizes, supplier coverage, credit limits, default rates, and disruptions. Assumptions and code should be published so others can reproduce the results.

Phase 3: Small, voluntary pilot

If the simulation and governance review support proceeding, run a limited opt-in pilot with a defined group of suppliers and users. Maintain appropriate existing accounting and payment arrangements during the pilot. Assess local legal, tax, employment, consumer-protection, and data obligations with qualified advisers before launch.

Proposed evaluation measures

Dimension	Example measures
Operational efficiency	Transaction cost; time to settle; failed or reversed transactions; hours spent reconciling accounts.
Adoption	Active participants; repeat use; share of eligible transactions settled in credits; supplier acceptance.
Credit health	Negative-balance exposure; arrears; defaults; concentration of positive and negative balances; redemption failures.
Price and liquidity	Exchange ratios; transaction spreads; price changes for a defined basket; unfilled offers.
Environmental outcomes	Absolute and per-service energy, material, water, waste, and greenhouse-gas impacts against a stated baseline.
Equity and access	Access to essential goods; participation barriers; distribution of benefits and losses; complaints and appeals.
Security and resilience	Data errors; disputes; privacy incidents; outage recovery; ability to transact during connectivity failures.

> The proposed architecture diagram is the only graph included here because Project World has no pilot data to graph. Numerical “efficiency” curves or forecasts would be invented. Any later performance figures should be based on disclosed methods and observed data.

11 Limitations

This is a conceptual design, not a tested monetary system. It does not specify a jurisdiction, member population, legal structure, final credit unit, software stack, or independent verification provider. It contains no transaction data, pilot results, or calibrated economic model.

The proposal also cannot assume that a local credit network would be independent of the wider economy. Participants may still rely on external suppliers, infrastructure, public services, and legal institutions. The degree of independence should be measured by tracking which inputs, expenses, and obligations remain outside the network.

Finally, resource measurement does not determine social value. A community still needs legitimate processes for deciding what to measure, how to handle scarcity, how to distribute essential goods, and who bears losses when commitments fail.

12 Conclusion

Project World is best treated as a researchable proposal for a voluntary, non-fiat credit-clearing network with explicit resource accounting — not as a proven replacement for money or the global economy. Its most useful contribution may be the separation of three questions that earlier versions combined:

1. What physical resources or services are available?
2. What claims or credit balances can participants hold and transfer?
3. What environmental and social outcomes result from the system?

Those questions require different records, rules, and evidence. A small, transparent, opt-in pilot could test whether the design improves exchange or resource visibility. It could also show where the system fails. Until such evidence exists, claims about sustainability, equity, inflation, or conflict should remain hypotheses.

References

[1] Bank for International Settlements. (2026). “Anchoring trust in money: innovation beyond stablecoins.” Annual Economic Report 2026, Chapter III.

[2] Fare, M., & Ould Ahmed, P. (2017). “Complementary Currency Systems and their Ability to Support Economic and Social Changes.” Development and Change, 48(5), 847–872. doi:10.1111/dech.12322.

[3] Hayek, F. A. (1945). “The Use of Knowledge in Society.” The American Economic Review, 35(4), 519–530.

[4] Albizri, A., & Appelbaum, D. (2021). “Trust but Verify: The Oracle Paradox of Blockchain Smart Contracts.” Journal of Information Systems, 35(2), 1–16. doi:10.2308/ISYS-19-024.

[5] United Nations Environment Programme. (2024). Global Resources Outlook 2024: Bend the Trend — Pathways to a Liveable Planet as Resource Use Spikes. International Resource Panel. ISBN 978-92-807-4128-5.

[6] Stamm, C. B. (2021). “Understanding the Recent Dynamics of Local Currency Initiatives in Switzerland.” International Journal of Community Currency Research, 25(2). doi:10.15133/j.ijccr.2021.0015.

[7] Olk, C. (2025). “Special on Purpose: Complementary Currencies in the Hierarchy of Money.” Sustainability Science, 20, 1155–1167. doi:10.1007/s11625-023-01444-8.

[8] Schraven, J. (2001). “Mutual Credit Systems and the Commons Problem: Why Community Currency Systems such as LETS Need Not Collapse Under Opportunistic Behaviour.” International Journal of Community Currency Research, 5. doi:10.15133/j.ijccr.2001.002.

[9] United Nations Framework Convention on Climate Change. (2026). Article 6.4 Standard: Validation and Verification, A6.4-STAN-AC-003, version 01.0(g).